

PROBLEM STATEMENT :

You are a Product Manager on the Core Experience team at Google Photos.

Over years of usage, users accumulate thousands of photos, videos, screenshots, documents, and other visual memories in Google Photos.

While users can easily search when they know *what* they are looking for, retrieval becomes much harder when memory is incomplete.

For example, a user may remember:

“That small café we went to during our Goa trip.” OR

“The picture of the medicine I took when I was sick last year.”

The user knows that the photo exists—but may not remember **when it was taken, where it was taken, what album it belongs to, or the exact words needed to search for it.**

One of your company’s strategic goals is to:

Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe when they start searching.

The challenge is **not to improve search in general.**

Your task is to understand how people remember old visual information, where the existing retrieval experience breaks down, and identify an opportunity that can meaningfully improve successful retrieval.

Part 1: Build an AI-Powered Discovery Engine

Before proposing any solution, build an AI-powered system that analyzes user feedback and conversations about photo retrieval at scale.

You may use:

- Claude
- GPTs
- Agents
- Workflows
- RAG
- n8n
- Zapier
- Perplexity
- Any AI-native stack of your choice

Analyze publicly available sources such as:

- Google Play Store reviews

- App Store reviews
- Reddit discussions
- Google Photos community/support discussions
- Social media conversations
- YouTube comments
- Forums and other relevant public discussions

Your discovery engine should help uncover questions like (sample questions only):

- *What kinds of old photos do users struggle to retrieve?*
- *What information do people actually remember about a photo?*
- *What information have they forgotten?*
- *How do users formulate searches when their memory is incomplete?*

Your workflow should go beyond summarizing reviews or performing sentiment analysis.

It should enable you to **identify and compare different retrieval problems and opportunity areas** using evidence from real users.

Part 2: Break Down the Business Metric

Break down:

Successful retrieval of vaguely remembered photos

into the relevant user behaviors and product outcomes.

Develop your own decomposition based on your understanding of the product and the evidence surfaced by your discovery engine.

Consider questions such as (sample questions):

- *Is the user unable to express what they remember?*
- *Does Google Photos fail to understand the clues they provide?*
- *Are potentially relevant results difficult to evaluate?*
- *Does the user struggle to refine an unsuccessful search?*

Use this decomposition to identify where the greatest opportunities may exist.

Part 3: Validate the Opportunity Through User Research

AI-generated insights are only a starting point.

Conduct **5–6 user interviews** with respondents from the target segment you choose.

Think carefully about the user research methodology you would choose.

Part 4: Define the Problem

Based on your discovery engine and primary research, clearly articulate:

- Your target user segment
- The retrieval scenario you are solving for
- The product outcome you intend to influence
- The root cause of retrieval failure
- Existing user workarounds
- Why solving the problem creates meaningful user value
- Why solving the problem makes business sense for Google Photos

Show how your thinking evolved across:

Business Metric → Product Outcomes → AI-Powered Discovery → Observed User Behavior → Problem Definition

DO NOT frame the problem simply as:

“Users find it difficult to search for old photos.”

Your research should uncover **why retrieval fails despite users retaining some memory of the photo.**

Part 5: Build an AI-Native MVP

Based on the problem you identify, build and deploy a functional AI-native MVP.

The MVP may take the form of:

- A feature within Google Photos
- An AI-powered retrieval workflow
- A conversational or multimodal retrieval experience
- An agent
- A standalone prototype

The MVP must be sufficiently functional that another person can use it to attempt a retrieval task.

Your research should determine **where intelligence is actually needed in the retrieval journey.**

Part 6: Test Your MVP With Users

Return to at least **3 users from your target segment** and ask them to interact with your MVP.

Where feasible, test it against real or representative retrieval tasks uncovered during your initial research.

Document what you learned and what you would change in the next iteration.

Part 7: Define Success

Define appropriate leading and diagnostic metrics for your solution.

Your final metric framework should reflect the solution you actually build.

Part 8: Risks & Mitigation Steps

Think about why your solution might fail.

Identify the most important risks for **your specific solution** and propose mitigation plans.

Deliverables

[Link] AI-Powered Discovery Engine

- Link where the workflow can be tested
- A 1-slide explanation inside the final deck outlining how the workflow works

[PDF] 10-slide deck

The deck should communicate:

- Business metric decomposition
- Discovery-engine findings
- User research and observed retrieval tasks
- Chosen target segment
- Root cause
- Problem definition
- Solution rationale
- MVP and user testing
- Success metrics
- Risks and limitations

[Link] Deployed AI-Native MVP

A publicly accessible prototype, workflow, or agent that can be interacted with and tested.

Guidelines for the deck

- Name of the Fellow should NOT be present anywhere in the slide deck
- **10 slides max** (title slide, if you choose to have it, should be counted within the 10 slides)
- Slide title should state the key message of the slide for easier reading. For example: don't write "Problem" as the slide title, state the problem succinctly in the slide title.
- If you are using a background colour for the slides, please ensure the text is readable.
- When using colours, keep in mind the reader may be colour blind. So, choose colours carefully.
- Link supporting artefacts (eg. survey URLs etc) via a hyperlink. Please ensure the reader has access to the documents. If you do not provide access, this might lead to reduction in your scores.
- Maximum file size should be less than **40 MB**
- Naming of the file should be **e.g. NL_PhonePe**
- Minimum font size: 14 for Google Slides or PPT (**this has to be strictly adhered to**) Minimum font size: 26 for Figma with frame of 1920*1080px (**this has to be strictly adhered to**)
- Minimum font size: 22 for Canva with frame of 1920*1080px (**this has to be strictly adhered to**)

Submission Deadline: 7 October, 2026 3:59:00 PM IST (no submissions will be accepted after the deadline, even if it is by a few seconds!)